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Main Idea: Finding a Safer Way to Store Calculators to Prevent Theft

Element A

The person involved in this problem is our community partner which is Mr. Scott. He is also our stakeholder along with Zachary Simmons, Richard Johnson, Frank Calhoun, and Immanuel Korkoyah. Our goal is to make a calculator holder for him. Since he is a math teacher, it is important for him to stay organized and know where everything is at all times. To help us make the best product possible, he is giving us constraints for what to do and what not to do. For example, he wants us to use recyclable materials that are also durable. This way, the calculator holder will be long-lasting and environmentally friendly. According to *Recycling-101*, “plastic bottles, beverage cans, paper, cardboard, food containers, and glass bottles” can be recycled as long as they are clean. On the other hand, “plastic bags, plastic wrap, flexible packaging, and polystyrene foam” are not recyclable. Therefore, we will avoid using those types of materials when constructing our product.

Our community partner often has to deal with students taking his calculators and taking them home. He has already had a few of his calculators taken. This often leads to a shortage of calculators for everyone. This has become such an issue he has stopped lending out calculators at all and is now making people get their own. He needs a way to help keep track of his calculators so that he knows how many calculators he has for better security.

The problem is happening not only in Roosevelt, but in schools all across the nation. Our local community project could help teachers across the country. Since most school provided calculators are stolen or accidentally taken, the teachers end up having to pay for the calculators for their classes, which aren't cheap at all. These stolen calculators could lead to less support in class or students who also can't afford a calculator to have to buy one. This problem usually occurs when students can't afford their own calculators or when they want to resell the calculators. According to Katie Wiseman, the sell price for calculators “range anywhere from \$100 to \$170”, making the resale price possibly even greater than that. College and high school students are the ones who end up buying these resold calculators when they cannot find them in stores. Teachers and students struggle affording calculators daily, let alone finding them. These calculators need to be protected by something secure, durable, and safe.

Many teachers find themselves wondering where their calculators are and where to store them because of two larger issues, Teacher burnout and a lack of school funding for storage. According to School of Education (2021), Teacher Burnout is a result of “prolonged exposure to poorly managed emotional and interpersonal job stress.” These stressors cause teachers constant fatigue, and in turn results in the misplacement of essential utilities such as computers, writing utensils, paperwork, and costly calculators which are simply too expensive to lose. Also,

teachers usually cannot handle the financial burden of paying out of pocket with teachers paying on average \$673 per year on classroom supplies according to a survey conducted by the Association of American Educators. These two problems happen everyday for educators, including Mr. Scott, so many teachers would benefit from a place to store their calculators and expand their teaching capabilities.

The goal of our design is to not only protect calculators, but to also help the community. School is much harder without a calculator to use, especially for highschool and college students. We want to make sure that this problem occurs less often or not even at all. Teachers and students both need these items, and to take the calculators just for monetary means is very selfish and foolish. Teachers can benefit from the holders by being able to keep better track of what happens to their calculators. This will make the classroom environment run smoother since teachers wont have stress about any misplacement, which results in the students understanding the learning material better.

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